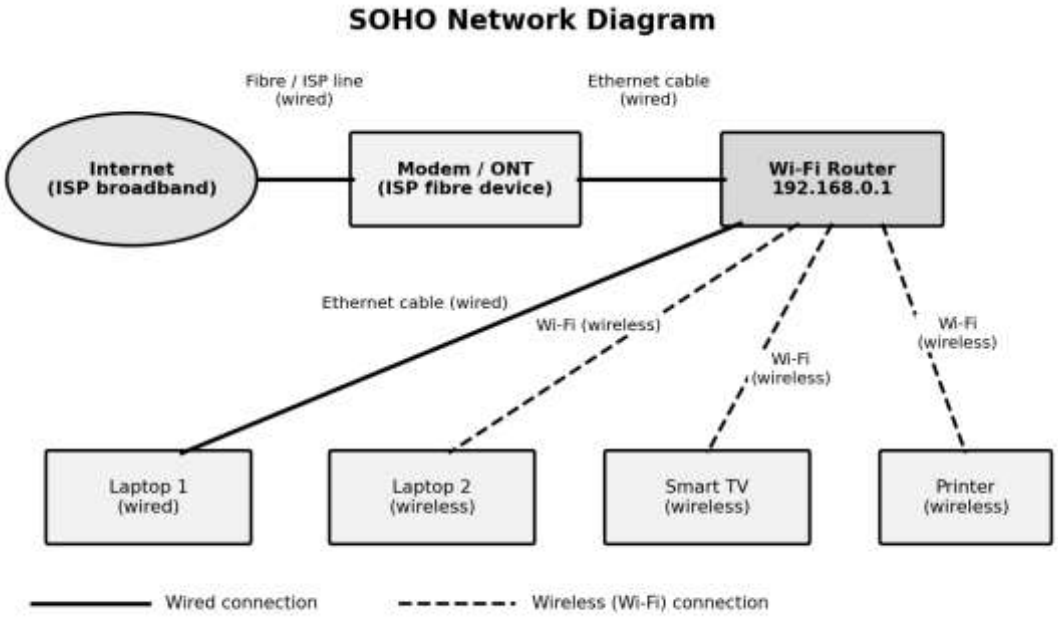
Build a SOHO Network

Assignment Answers

Q1. Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).



Connection	Type	Description
Internet to Modem / ONT	Wired (fibre)	The ISP's broadband connection (for example Jio Fiber, Airtel Xstream or ACT) enters the office through the ISP's modem/ONT.
Modem / ONT to Wi-Fi Router	Wired (Ethernet cable)	A Cat6 patch cable goes from the ONT's LAN port to the router's WAN (internet) port.
Router to Laptop 1	Wired (Ethernet cable)	Laptop 1 is connected to a LAN port of the router with a Cat6 cable for a stable connection.
Router to Laptop 2	Wireless (Wi-Fi)	Laptop 2 connects to the router's 5 GHz (or 2.4 GHz) Wi-Fi.
Router to Smart TV	Wireless (Wi-Fi)	The TV streams videos over Wi-Fi.
Router to Printer	Wireless (Wi-Fi)	A Wi-Fi printer is shared by both laptops through the router.

Q2.

List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.

Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.

No.	Item	Suggested brand / model	Purpose	Qty
1	Broadband connection and modem/ONT	ISP plan such as Jio Fiber, Airtel Xstream Fiber or ACT; the ONT/modem is provided and installed by the ISP	Brings the internet into the office	1
2	Wi-Fi router	TP-Link Archer AX23 (AX1800 dual-band Wi-Fi 6, 4 Gigabit LAN ports, WPA3 support). Price seen on Indian listings: about Rs 3,300 to Rs 4,700, depending on seller and day	Shares the internet by cable and Wi-Fi, gives IP addresses (DHCP)	1
3	Laptops	The two laptops already owned (with Wi-Fi and an Ethernet port; use a USB-C to Ethernet adapter if there is no port)	Work devices	2
4	Smart TV	The existing Wi-Fi smart TV (for example Mi, Samsung or Sony)	Streaming and presentations	1
5	Wi-Fi printer	Canon PIXMA MegaTank G3000 (Rs 12,999 to Rs 13,749 on listings) or HP Smart Tank 589 (about Rs 13,699 in July 2026)	Wireless print, scan and copy for both laptops	1
6	Ethernet cables	Cat6 patch cables, 1 m and 2 m (brands such as D-Link, TP-Link or Belkin)	ONT to router and router to Laptop 1	2
7	Surge-protected power strip / small UPS (recommended)	Any 600 VA UPS (APC, Luminous, Microtek) or a surge-protected extension board	Protects the router and devices from power cuts and spikes	1
8	Network switch (optional)	TP-Link TL-SG105 (5-port Gigabit switch)	Only if more wired devices are added than the router's 4 LAN ports	0 or 1
9	Wi-Fi extender (optional)	TP-Link RE-series range extender	Only if the signal is weak in some rooms	0 or 1

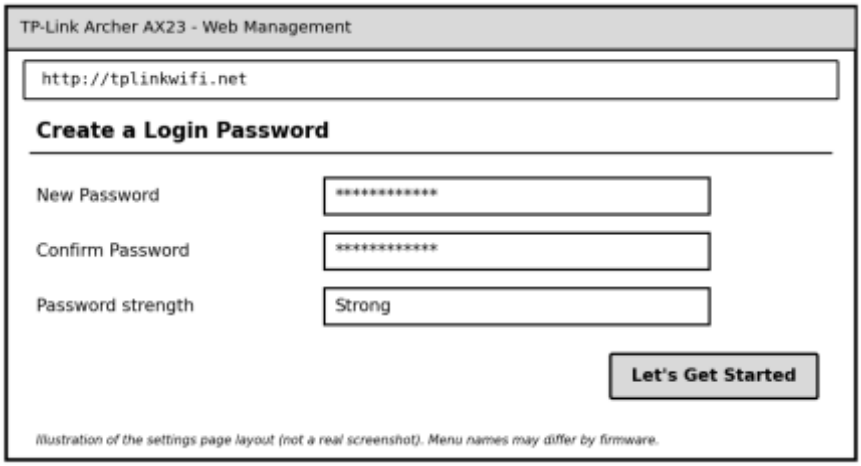
Note: A switch and an extender are not required for this small network, because the router has 4 LAN ports and one router covers a small office. Prices change often, so check the current listing on Amazon or Flipkart before buying.

Q3 Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

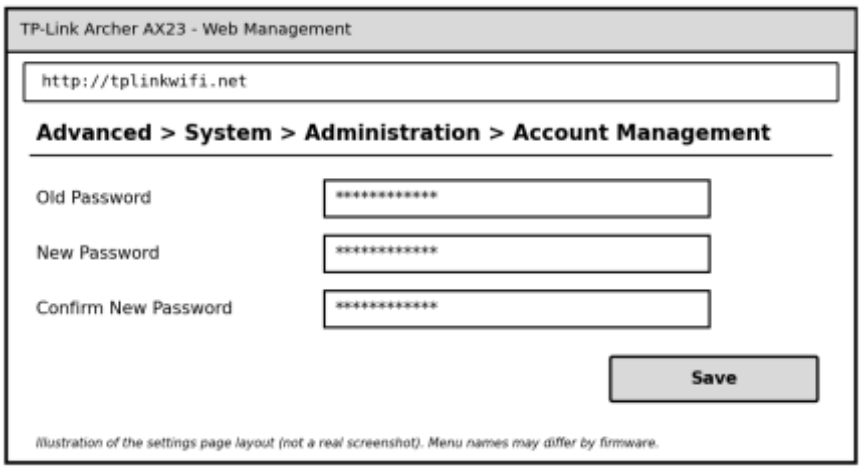
1. Enable WPA2 security: open Advanced, Wireless, Wireless Settings. Set Security to WPA2-PSK with AES encryption (WPA2/WPA3 is also acceptable if all devices support it). Never leave the network Open.
2. Change the admin password later if needed: Advanced, System, Administration, Account Management, enter the old password, then the new password twice and save.
3. Extra hardening: turn off WPS, turn off remote management, and check Advanced, System, Firmware Upgrade for new firmware.
4. Save the settings. The Wi-Fi restarts, so reconnect the laptops and the smart TV with the new SSID and password, and test the internet by opening a website.

Note: Menu names may differ a little between router brands and firmware versions.

Screenshots of my router configuration (login password, Wi-Fi name and WPA2 security, and admin password change):







Q4. Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

Network: 192.168.0.0/24, subnet mask 255.255.255.0, router (default gateway) 192.168.0.1. The DHCP pool of the router is 192.168.0.100 to 192.168.0.199. Static addresses are chosen outside this pool, so no address conflict can happen. The MAC addresses are made up.

Device	MAC address (made up)	IP address	Assignment	Connection
Wi-Fi Router (LAN)	A4:2B:B0:5C:11:01	192.168.0.1	Static (router itself)	-
Laptop 1	3C:52:82:7A:9D:11	192.168.0.101	DHCP	Wired
Laptop 2	F8:75:A4:2E:60:5B	192.168.0.102	DHCP	Wireless
Smart TV	64:CB:E9:3F:A2:17	192.168.0.30	Static	Wireless
Printer	18:CF:5E:B4:08:C3	192.168.0.20	Static	Wireless

Common settings for every device

- Subnet mask: 255.255.255.0. Default gateway: 192.168.0.1. DNS server: 192.168.0.1 (the router) or 8.8.8.8.

Laptops (DHCP)

- Windows: Settings, Network and Internet, Wi-Fi (or Ethernet), the network, IP assignment, Edit, choose Automatic (DHCP). The router gives an address from 192.168.0.100 to 192.168.0.199.

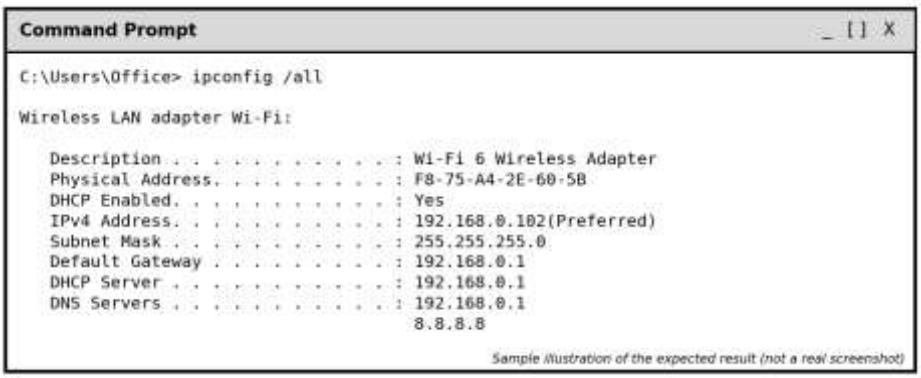
Printer and smart TV (static)

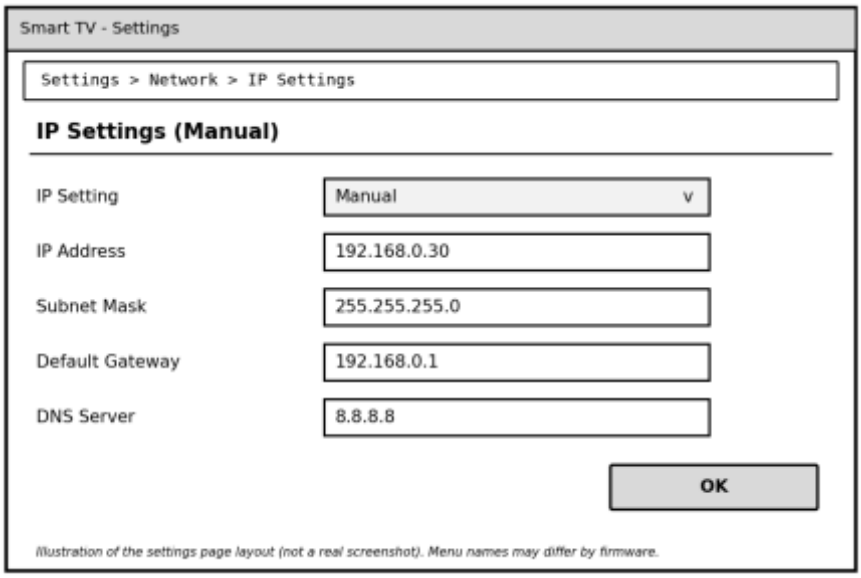
- Smart TV: Settings, Network, IP Settings, change to Manual, then enter IP 192.168.0.30, mask 255.255.255.0, gateway 192.168.0.1 and DNS 8.8.8.8.
- Printer: on the printer's control panel (or its web page) open Network Settings, TCP/IP, set Manual, then enter IP 192.168.0.20, mask 255.255.255.0 and gateway 192.168.0.1.
- Alternative: create a DHCP reservation in the router (Advanced, Network, DHCP Server, Address Reservation) that links each MAC address to its IP. The device then always gets the same address.

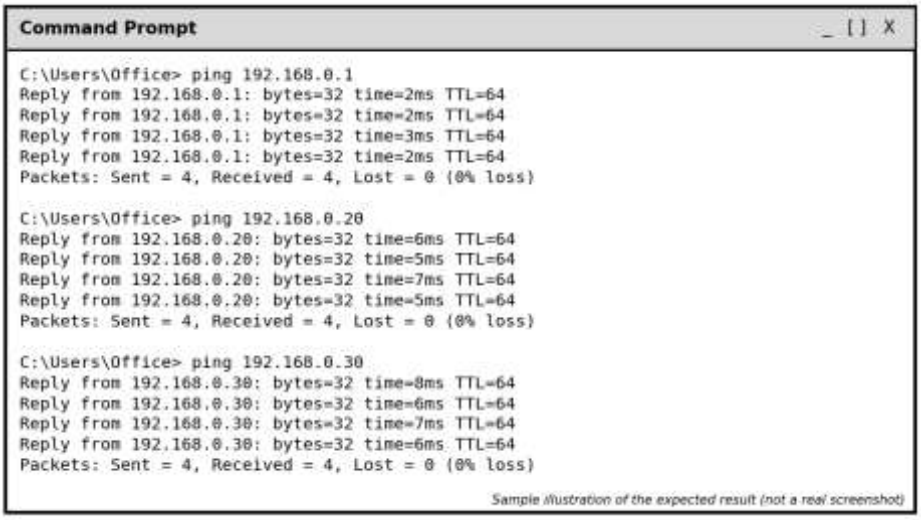
Verification

The commands ipconfig /all (to see the IP settings) and ping (to test the router, printer and smart TV) were used; their results are shown below.

Screenshots of my IP configuration:







Q5. Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

First check whether the problem is only on this laptop: if the other laptop, the TV and the phone work, then the router and the internet are fine and the fault is on the laptop. If all devices fail, check the router and the ISP.

Possible cause	Symptoms	Diagnosis steps	Fix
1. Wi-Fi is off, in Airplane mode, or the wrong password was used	No Wi-Fi icon or 'Not connected'; the network is not joined	Check the Wi-Fi switch and Airplane mode; check whether the SSID is visible	Turn Wi-Fi on, forget the network and connect again with the correct password
2. IP address problem (no valid address, or a conflict)	'Limited connectivity'; address starts with 169.254.x.x or a duplicate address warning	Run ipconfig /all and look at the IPv4 address and gateway	Run ipconfig /release then ipconfig /renew; set the adapter to 'Obtain IP automatically'; remove any wrong static address

Possible cause	Symptoms	Diagnosis steps	Fix
3. DNS problem	Pings to 8.8.8.8 work but websites do not open	ping 8.8.8.8 and then ping google.com; if the second fails it is a DNS fault	Run ipconfig /flushdns; set the DNS to 8.8.8.8 and 1.1.1.1 (or the router)
4. Weak signal or interference	Slow connection or frequent disconnection; low signal bars	Check the signal bars and compare with other devices in the same place	Move closer to the router, use the 5 GHz network, or add an extender
5. Router blocking the device (MAC filter / parental control) or router/ISP failure	Connected to Wi-Fi but no internet only for this laptop, or on all devices	Check the router's access control list; check the internet light on the router; test another device	Remove the laptop from the block list; restart the router and ONT; contact the ISP if the WAN light is off
6. Driver, VPN, proxy or firewall problem	Other devices work; the error appears after a software change	Run the Windows Network Troubleshooter; check VPN/proxy settings and Device Manager	Disable the VPN/proxy; update or reinstall the Wi-Fi driver; run netsh winsock reset and netsh int ip reset, then restart

Step-by-step test order

The laptop is tested in this order: ping 127.0.0.1 (laptop software), ping 192.168.0.1 (router), ping 8.8.8.8 (internet) and ping google.com (DNS). The result is shown below.

The first command that fails shows where the problem is: the laptop itself, the Wi-Fi link to the router, the router/ISP connection, or DNS.

Screenshot of my troubleshooting test:

